

Section 3.6 — Interpretation & Multi-Omics Analysis Report

1. Multi-Omics Layer Weighting Strategy

In our integration pipeline, an **equal-weights strategy** ($w_{\text{genomic}} = w_{\text{transcriptomic}} = w_{\text{proteomic}} = 1/3$) was implemented. While this serves as an unbiased baseline, compelling biological arguments exist both for maintaining equal weights and for reweighting layers based on causal proximity to disease etiology

Argument for Equal Weights (1/3, 1/3, 1/3)

- **Triangulation Without Bias:** Equal weighting treats all three biological strata as independent sources of evidence. Because each layer suffers from unique technological noise (e.g., MS/MS peptide missingness in proteomics, hybridization/probe artifacts in transcriptomics, or linkage disequilibrium tagging in GWAS), equal weighting minimizes bias toward any single experimental platform.
- **Balanced System Coverage:** It ensures that a candidate target requires support across multiple biological levels to reach the top quantiles, effectively penalizing single-layer false positives.

Argument for Up-Weighting Genomics (GWAS)

- **Establishing Causality vs. Confounding:** GWAS measures germline genetic variation fixed at conception, rendering it immune to reverse causation or confounding from secondary disease pathology. In contrast, transcriptomic and proteomic alterations in post-mortem Alzheimer's disease (AD) brains often reflect downstream cellular stress, neuroinflammation, or terminal neurodegeneration rather than primary drivers.
- **Proposed Reweighting:** Allocating higher weight to GWAS ($w_{\text{genomic}} = 0.50$, $w_{\text{transcriptomic}} = 0.25$, $w_{\text{proteomic}} = 0.25$) prioritizes targets with causal evidence, ensuring that observational expression changes only elevate targets with strong germline risk.

2. Top Target Recovery and Novel Candidates

Recovered Known AD Targets

The gene-level integration pipeline successfully recovered core benchmark risk loci and functional drivers associated with Alzheimer's disease:

Target Gene	Overall Rank (out of 1,317)	Biological Relevance in AD
CR1	11	Complement receptor 1; key regulator of neuroinflammatory cascades and immune-mediated clearance.
MAPT	14	Microtubule-associated protein tau; primary constituent of neurofibrillary tangles.
TREM2	20	Triggering receptor expressed on myeloid cells 2; central driver of microglial activation and amyloid plaque phagocytosis.
APOE	34	Apolipoprotein E; dominant genetic risk factor ($\epsilon 4$ allele) involved in lipid transport and amyloid- β clearance.
PICALM	50	Phosphatidylinositol binding clathrin assembly protein; involved in clathrin-mediated endocytosis and $A\beta$ clearance.
CLU	82	Clusterin / Apolipoprotein J; extracellular chaperone suppressing amyloid aggregation

Non-Obvious Hits Worth Second-Look Validation

- **SAMHD1 (Rank 7):** Exhibits strong germline association ($-\log_{10} P = 35.70$) paired with robust concordant upregulation across RNA ($\log_2 FC = 2.43$) and protein ($\log_2 FC = 0.25$). SAMHD1 is an innate immune regulator involved in dNTP hydrolysis and viral restriction; its upregulation in AD post-mortem tissue suggests an active neuroinflammatory or cGAS-STING-driven innate immune response in glial cells.
- **CD2AP (Rank 2) & PECAM1 (Rank 4):** Both candidates demonstrate top-decile multi-evidence scores driven by genome-wide significance and pronounced expression shifts, highlighting vascular-immune dysfunction in AD pathogenesis.

3. Analysis of a Discordant Gene (concordant == False)

Selected Candidate: CD2AP (Rank 2)

- **RNA Log2 Fold-Change:** -3.38 (strong downregulation)
- **Protein Log2 Fold-Change:** $+0.22$ (upregulated)
- **GWAS Significance** ($-\log_{10} P$): 21.10
- **Concordance Status:** False

RNA Expression (Downregulated: -3.38)

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Post-Transcriptional / Translational Buffering

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Protein Accumulation (Upregulated: +0.22)

4. Methodological Limitations: Unmatched Gene-Level Integration vs. Matched Sample-Level Integration

Unmatched Gene-Level Integration (Part 3 Approach)

- **Methodology:** Merges population-averaged summary metrics (e.g., max GWAS p -values, mean cohort differential expression $\log_2 FC$) across distinct study cohorts lacking overlapping individuals.
- **Limitation:** Prevents sample-wise covariance or correlation analysis (e.g., Spearman correlation between RNA and protein within individual patients).
- **Clinical Boundary:** Results represent **population-level disease priors**. This framework cannot evaluate patient heterogeneity, infer individual disease subtypes, or support per-patient diagnostic or therapeutic claims.